

Getting Started with the Oracle AI Data Platform

The **Oracle AI Data Platform** empowers organisations to unlock the full potential of their data with a modern, AI-ready foundation. Designed for speed, simplicity, and scale, it unifies data lake storage, cataloging, ingestion, preparation, and analysis into one seamless experience—enabling teams to move from raw data to real insights faster than ever.

With just a few clicks, users can create a secure AI data environment, build rich catalogs across their entire data lake and data estate, and manage access using enterprise-grade RBAC. Built-in Spark-powered notebooks make it easy to explore, transform, and enrich data, while automatic metadata ingestion from supported services keeps everything organised and up to date.

Because Oracle Analytics Cloud (OAC) integrates directly with the platform, business users can instantly analyse curated datasets without complexity or delays. The result is a single, unified “pane of glass” that streamlines the entire data and AI lifecycle—backed by Oracle’s proven security and reliability.

Core Capabilities of Oracle AI Data Platform

Data Integration - Seamless access to any data source—live or ingested—backed by deep integration with Oracle technologies and applications, and enhanced by zero-ETL data sharing from Fusion and other Oracle applications..

Unified Catalog -A Unified master catalog that governs both data and AI assets, supporting the full medallion architecture—from bronze to silver to gold. This enables fast ingestion, curation, and delivery of data products, AI applications, and ML models across diverse formats, all with governance built in at every layer.

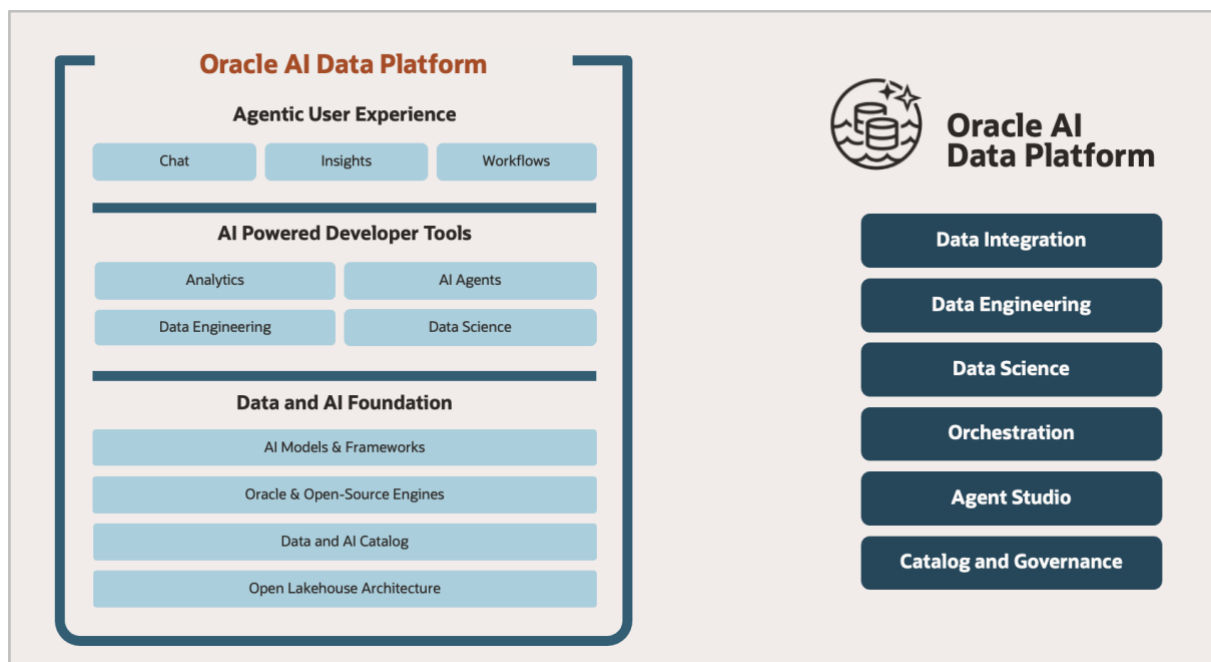
Data Governance- Data governance in the Oracle AI Data Platform ensures secure, well-managed, and organized data across all catalogs. With RBAC access control, metadata management, and audit features, teams can maintain data quality and confidently run their AI and analytics workloads.

Data Engineering & Data Science -AI Data Platform provides a unified development experience for both data engineers and data scientists, with flexible CPU/GPU compute to match any workload. Developers can work in their preferred languages—from Python and Spark SQL supported by seamless, unified workflow orchestration.

AI Models and Framework: The AI Data Platform seamlessly integrates with Oracle’s Generative AI services, supporting leading foundational LLMs like Cohere, OpenAI, and Llama. It also works effortlessly with open-source frameworks—including PyTorch,

TensorFlow, and LangChain—providing full flexibility to build and deploy AI models in any architecture.

AI Agents & Application :-AI Data Platform supports pipelines that incorporate AI and GenAI by accessing both built-in and external models. For dedicated AI development—such as building agents—the platform offers both low-code and code-first experiences, powered by OCI Gen AI and Oracle 23ai, including its vector store. Developers can manage the full lifecycle of AI assets from development through production, and these agents or tools can even be integrated into Fusion AI Agent Studio to extend the Fusion experience.



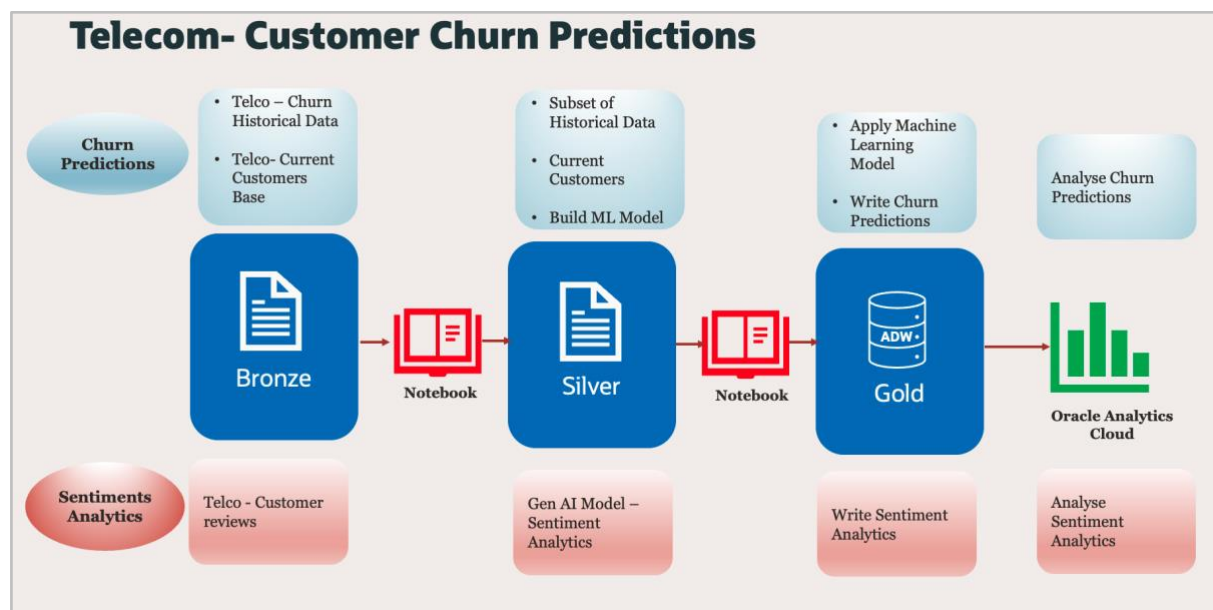
How the AI Data Platform Delivers Value Through the Medallion Architecture

To showcase the power of the AI Data Platform and its Medallion Architecture, let's explore a telecom customer churn use case. This example demonstrates how data flows from raw ingestion to refined analytics and AI-driven insights.

Objective

The objective of this use case is to demonstrate how the AI Data Platform, using the Medallion Architecture, predicts customer churn and leverages an AI Model to analyze customer sentiment, uncover issues, and improve retention.

How We Built This Use Case



In this use case, we demonstrate how the AI Data Platform, built on the Medallion Architecture, can drive actionable insights for telecom customer retention.

We start by ingesting both historical churn data and current customer information into the Bronze layer. This data is then refined in the Silver layer, where we also train a machine learning model to predict customer churn. The model's predictions for current customers are stored in the Gold layer within an Autonomous Database and visualized in Oracle Analytics Cloud to provide clear, actionable insights.

In the second part of the use case, we analyze customer sentiment from unstructured reviews. The review data is refined in the Silver layer and processed using a Generative AI LLM to extract sentiment and highlight key topics. These insights are then saved in the Gold layer and visualized in Oracle Analytics Cloud, helping identify customer pain points and areas for improvement.

AI Data Platform Catalogs

As part of the Medallion Architecture, we set up three catalogs—Bronze, Silver, and Gold—within the AI Data Platform, providing a structured foundation for data processing, analytics, and AI-driven insights.

Bronze Catalog – Raw Data

The Bronze catalog stores all raw, unprocessed data in object storage. For this use case, it includes 10 years of historical customer churn data as well as current customer data. This catalog serves as the foundation for all further processing.

Silver Catalog – Refined Data and Machine Learning

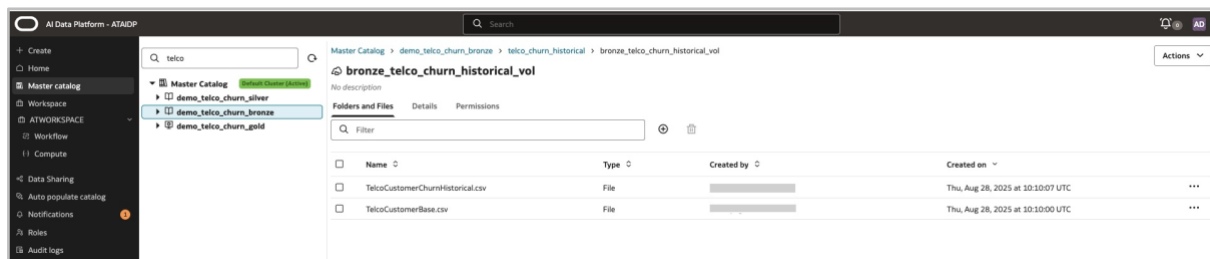
In the Silver catalog, we refine the raw data from Bronze. This involves cleaning, subsetting, and transforming the data, which we then store as Parquet files in object storage. Using the historical churn data in this catalog, we also train a machine learning model for churn prediction, which is saved in Silver for later use.

Gold Catalog – Analytics-Ready Data

The Gold catalog is the consumption-ready catalog backed by an Autonomous Database for analytics and reporting. Here, we apply the churn prediction model to the current customer data, calculate the churn probabilities, and store the results in an ADW table for visualization and insights.

Sentiment Analysis – From Raw Reviews to Actionable Insights

For the sentiment analysis part, unprocessed customer reviews are first stored in the Bronze catalog. After refinement in Silver, we run an AI Model to extract sentiment—positive, negative, or neutral—as well as key topics or aspects. These insights are then stored in the Gold catalog as a database table, ready for analytics in Oracle Analytics Cloud to uncover customer pain points and improvement areas.



AI Data Platform Notebooks

We developed a set of Python notebooks to move data across catalogs, build the ML model, and perform sentiment analysis using an AI Model.

Notebook 1 – Process Historical Churn Data

- Reads historical churn data from the Bronze catalog.
- Filters it to the most recent four years and selects relevant attributes.
- Writes the processed dataset into the Silver catalog.

Notebook 2 – Process Current Customer Data

- Reads current customer data from the Bronze catalog.
- Performs similar cleaning and transformation steps.

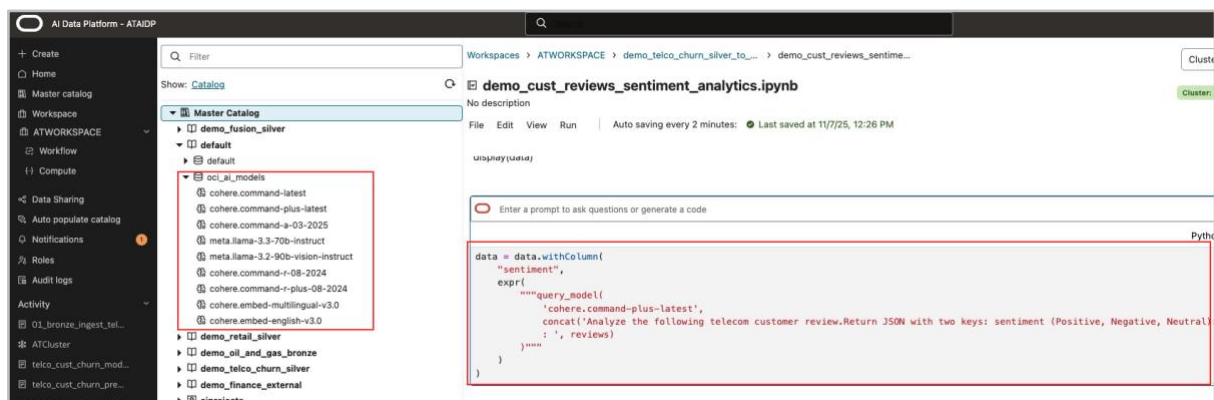
- Writes the processed dataset into the Silver catalog.

Notebook 3 – Train and Apply Churn Prediction Model

- Trains a churn prediction model using the historical dataset in the Silver catalog.
- Applies the trained model to current customer data to generate churn risk predictions.
- Stores the prediction results in the Gold catalog for analytics and reporting.

Notebook 4 – Sentiment Analysis on Customer Reviews

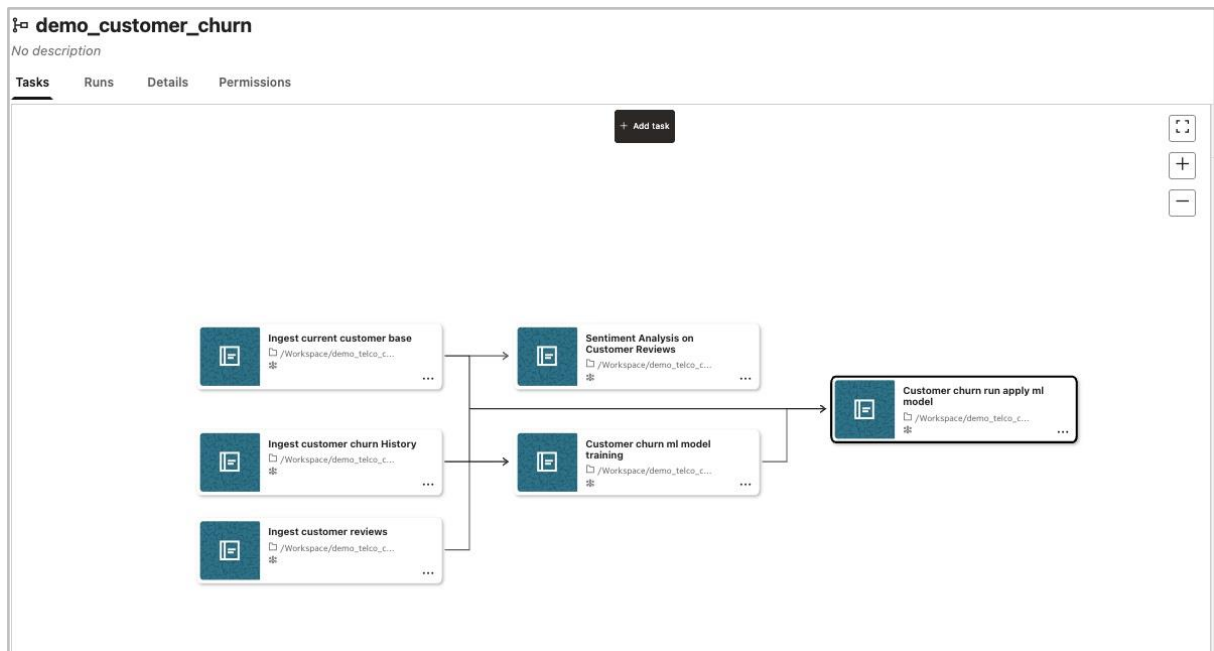
- Uses AI Model(Cohere) to process customer review text.
- Extracts sentiment scores (positive, negative, neutral) and key topics/aspects.
- Writes the enriched sentiment results to the Gold catalog for further analysis and insights.



AI Data Platform Workflow

Workflows in the Oracle AI Data Platform enable users to automate and orchestrate complex data pipelines. They can run on-demand or on a schedule and consist of multiple tasks with features like dependencies, triggers, and error handling.

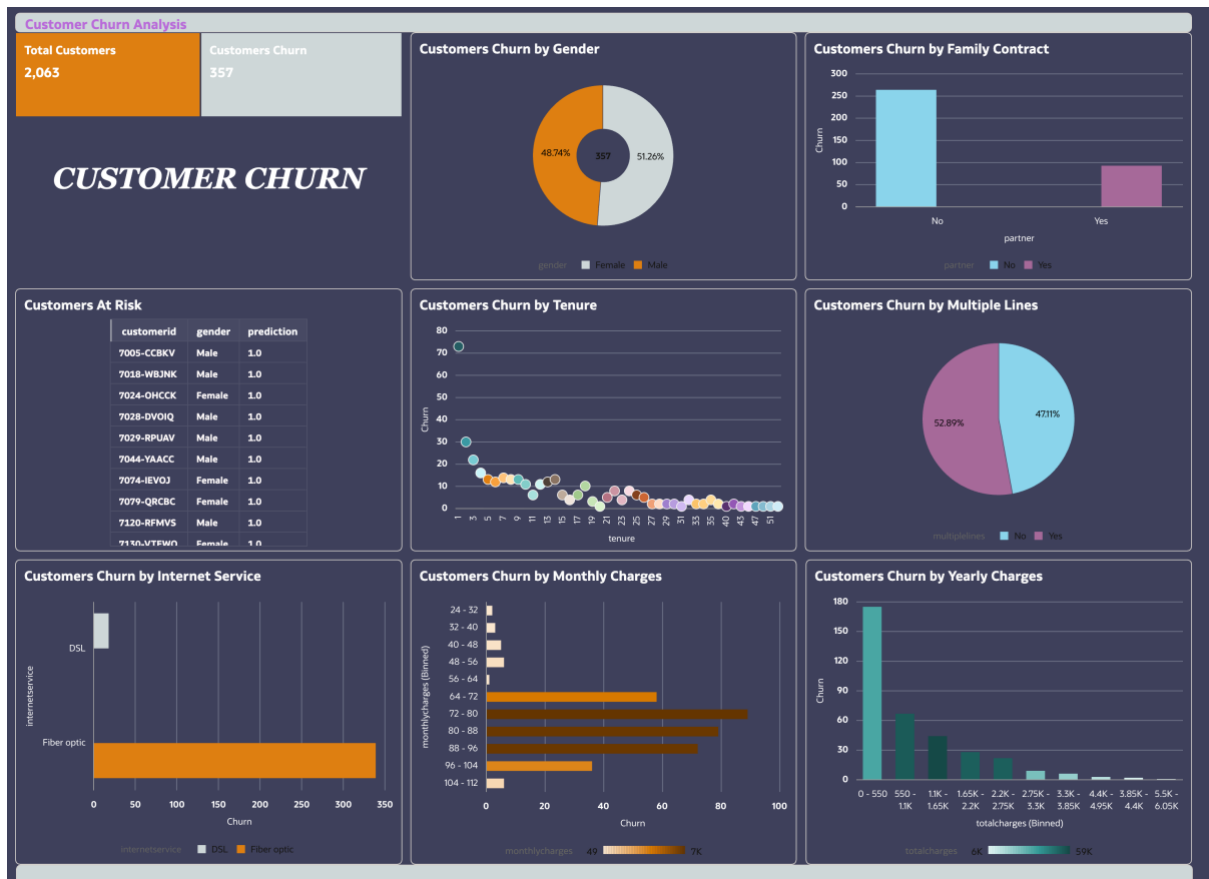
We created a workflow that orchestrates all notebooks, covering every task from data ingestion to ML model predictions and sentiment analysis. When executed, this workflow runs the entire process end-to-end automatically.



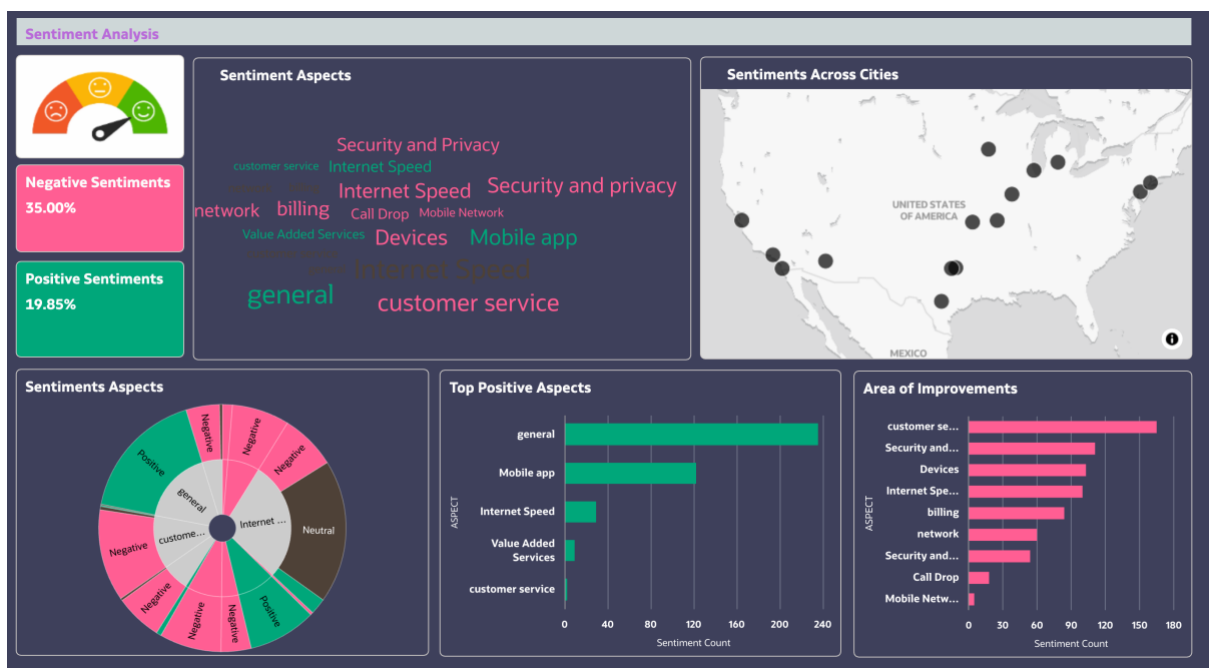
Explore Gold Catalog Insights with Oracle Analytics Cloud

Oracle Analytics Cloud provides a direct connector for the AI Data Platform, allowing you to access catalog data and analyze it directly through the Oracle Analytics Cloud workbook.

We connected Oracle Analytics Cloud to the Gold catalog, enabling us to explore churn predictions and sentiment analysis results interactively within an Analytics Cloud workbook, turning raw data into actionable insights.



The first dashboard lets us see which customers are at risk of churn according to our ML model. By exploring patterns across gender, contract type, tenure, and billing details, we can uncover the factors driving churn and pinpoint strategies to retain customers.



In the second dashboard canvas, we explore sentiment insights from customer reviews. Negative aspects, such as network quality or customer service, are highlighted in red, while positive aspects like mobile app features or value-added services appear in green. We also compare sentiments across cities to uncover regional trends, highlighting strengths and opportunities for improvement.

To conclude, this use case demonstrates how the Oracle AI Data Platform, leveraging the Medallion Architecture, unifies data engineering, machine learning, and AI-driven analytics to deliver actionable customer insights and drive smarter retention strategies.

Oracle AI Data Platform References

To explore more about the Oracle AI Data Platform, please refer to the official documentation and GitHub page.

[Oracle AI Data Platform Documentation](#)

[Oracle AI Data Platform Github Repository](#)